

Conditional Generative Modeling of Interception-Movement Fingerprints

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Abstract

We study whether fast planar interception movements admit compact participant-specific latent representations that can generate held-out distributions of trajectory shape and timing. The confirmatory analysis retains the interval from target-motion onset to finger arrival, filters the table-plane trajectory at 10 Hz, and resamples position to 100 phase points while predicting physical initiation and movement time separately. A conditional variational autoencoder (CVAE; $n=2,3,4,8$) is evaluated against spline+PCA, a deterministic conditional autoencoder, an unconditional VAE, and a task-condition-only stochastic Ridge baseline. Four participant folds provide out-of-fold predictions for all 28 participants; three neural seeds assess optimization variability. Spline+PCA is strongest for deterministic reconstruction, whereas CVAE $n=3$ and $n=8$ improve timing and generated-distribution fidelity. The $n=8$ model is the stronger generator; $n=3$ remains the compact interpretation target. Unconditional VAE results are statistically comparable, so participant-specific effects of explicit task conditioning are not established. Minimum-jerk component fidelity is modest and sensitive to model-order assumptions.

Primary conclusion: the evidence supports a capacity-interpretability tradeoff, not a claim that the CVAE is uniformly more accurate than classical or neural comparators.

1. Motivation and research question

Interception combines feedforward prediction with online correction [1]. Individual motor signatures have been demonstrated in other movement paradigms [2], but an interception model must additionally reproduce stochastic within-person variation. We therefore ask: **can a low-dimensional generative fingerprint, enrolled from a subset of a previously unseen participant's trials, reproduce the distribution of that participant's remaining movements?** This formulation separates participant enrollment from zero-shot prediction and treats full movement distributions, rather than single trajectories, as the target.

Contributions

(i) leakage-resistant participant rotation and disjoint context/query evaluation; (ii) matched deterministic, unconditional, condition-only, and spline baselines; (iii) explicit separation of phase-normalized shape from physical timing; and (iv) a secondary minimum-jerk decomposition derived from Prof. Friedman's public implementation.

2. Dataset and preprocessing

The source dataset contains 240 Hz electromagnetic finger-position recordings. We use 4,732 retained free-eye-movement trials (condition 2) from 28 participants. Filename metadata provides starting-position/speed category, stimulus side, and repetition; MAT metadata supplies outcome labels and exact target speed. The unreliable receive-time column is ignored and timing is derived from the frame counter. The primary analysis uses lateral x and forward y because the task is performed on a tabletop; out-of-plane z is excluded.

Each trajectory begins at target-motion onset (marker 5) and ends at recorded finger arrival. This preserves the pre-movement waiting interval. Position is low-pass filtered with a fourth-order 10 Hz Butterworth filter, translated to the trial origin, and resampled to 100 phase points. Resampling removes physical duration; therefore initiation time and movement time are withheld from the encoder and predicted by separate decoder heads. Trials missing required events or not reaching the interception site are excluded by the canonical audit. The late-trial rule and the condition-2 fixation message remain task-specific assumptions for Prof. Friedman's review.



Four outer participant folds; each participant tested once; neural seeds 42/43/44

Figure 1. Frozen strategy-window protocol. Outer folds contain 17 train, 4 validation, and 7 test participants; every participant is tested exactly once. Within each test participant, context trials estimate the fingerprint and disjoint query trials define empirical target distributions.

2.1 Inputs and targets

Element	Representation	Role
Trajectory	100 x 2 phase-normalized x-y positions	encoder input and decoder output
Task condition	sp category, side, exact executed target speed	CVAE/CAE condition vector
Physical timing	initiation time; movement time	decoder targets only
Fingerprint	mean context posterior coordinate	participant enrollment for generation

2.2 Feasibility check

A pre-specified 28-cluster K-means probe produced low absolute agreement (trajectory ARI=0.052; feature ARI=0.070) but exceeded its permutation null in both representations ($p=0.0050$). This is evidence of weak non-random participant structure, not evidence of 28 clean clusters.

3. Models and training

The CVAE follows the stochastic latent-variable formulation of Kingma and Welling [3] and the conditional structured-output extension of Sohn et al. [4]. The encoder maps trajectory and task condition to diagonal-Gaussian posterior parameters; the decoder reconstructs trajectory and log-transformed timing. Training minimizes trajectory reconstruction loss, weighted timing loss, and KL divergence to the standard-normal prior. Participant-balanced sampling prevents high-trial-count participants from dominating batches. Early stopping is selected on validation participants only.

Comparator	Question isolated
Spline+PCA (n=2,3,4,8)	classical low-rank shape reconstruction and stochastic coefficient generation
Conditional AE (n=3,8)	effect of KL regularization/stochastic bottleneck
Unconditional VAE (n=3,8)	incremental value of explicit task conditioning
Condition-only Ridge	performance available from task metadata without a participant fingerprint

4. Evaluation

Observed-trial reconstruction uses posterior means. Participant-balanced trajectory MSE and timing MAE summarize out-of-fold prediction. Closed-set enrollment assigns query trials to the nearest context fingerprint among the seven participants in that fold; nominal chance is 1/7. Generative fidelity compares context-conditioned samples with query trials using per-feature two-sample KS statistics across 11 active 2-D kinematic features, RBF-MMD, and energy distance. Lower values indicate closer empirical/generated distributions. Model contrasts use two-sided paired Wilcoxon tests on 28 seed-averaged held-out participants with Benjamini-Hochberg correction. Fold effects and optimization seeds are retained separately.

4.1 Minimum-jerk secondary analysis

Recorded and generated movement velocities are decomposed into one to four overlapping 2-D minimum-jerk components [5] using Prof. Friedman's implementation (repository commit 9c2f40c). Each component has onset, duration, lateral displacement, and forward displacement. Our primary order rule selects the smallest model with normalized fit error ≤ 0.05 , then ≤ 0.10 , otherwise the minimum-error candidate. Because the repository fits candidate orders but does not prescribe order selection, BIC and parameter sensitivities are reported explicitly.

4.2 Reproducibility

The confirmatory protocol is frozen as strategy-confirmatory-v1. The complete matrix contains 48 CVAE, 24 conditional-AE, 24 unconditional-VAE, 16 spline+PCA, and 12 condition-only runs. Neural seeds 42/43/44 vary initialization and training randomness; four outer rotations vary the held-out cohort. All metrics are derived from out-of-fold predictions only.

5. Benchmark results

Figure 2. Out-of-fold benchmark endpoints (28 held-out participants)

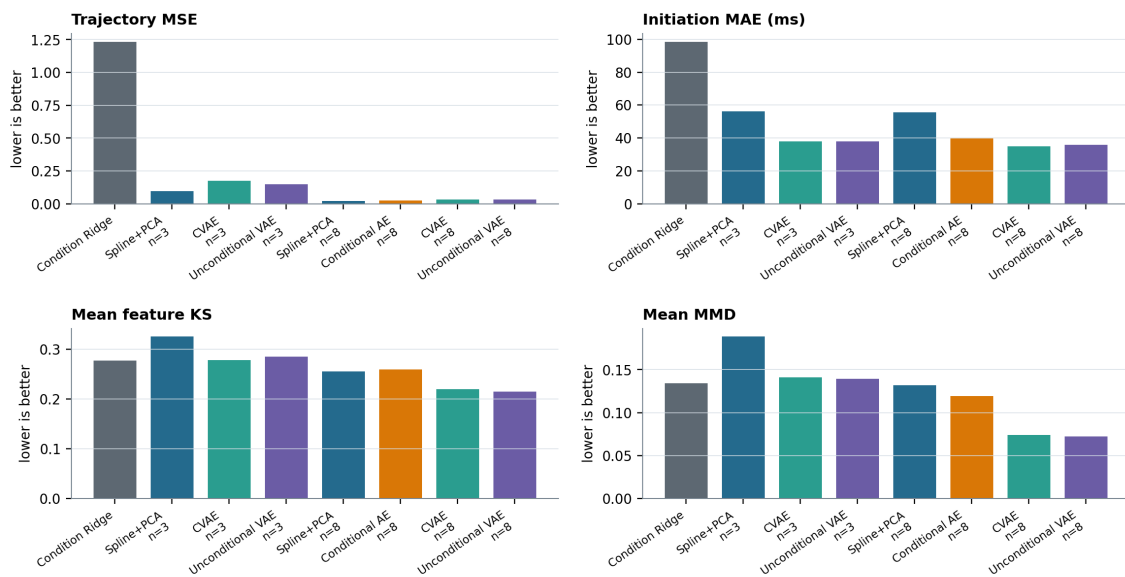


Figure 2. Participant-balanced out-of-fold endpoints. Spline+PCA minimizes deterministic trajectory MSE. Neural models provide lower timing error and, at $n=8$, lower generated-distribution distances. Metrics are not interchangeable; no single row is globally dominant.

Model	n	Traj. MSE	Init. MAE ms	Move MAE ms	Mean KS	MMD	Enroll. acc.
Condition Ridge	-	1.234	98.5	72.7	0.277	0.134	-
Spline+PCA	3	0.097	56.2	51.2	0.325	0.189	36.4%
CVAE	3	0.177	37.8	43.4	0.278	0.141	40.9%
Unconditional VAE	3	0.148	37.9	42.9	0.285	0.140	40.0%
Spline+PCA	8	0.023	55.6	50.2	0.255	0.132	48.6%
Conditional AE	8	0.026	39.9	42.7	0.259	0.119	46.5%
CVAE	8	0.032	34.8	44.2	0.219	0.074	50.7%
Unconditional VAE	8	0.033	36.0	42.2	0.215	0.072	50.8%

Table 1. Aggregate out-of-fold metrics. Enrollment accuracy is unavailable for the condition-only baseline. The $n=8$ CVAE reaches 50.7% closed-set enrollment versus 14.3% nominal chance, but this setting assumes context trials from each candidate test participant and is not zero-shot identification.

5.1 CVAE versus the spline baseline

The central comparison is a tradeoff. At $n=3$ and $n=8$, spline+PCA reconstructs observed trajectories more accurately, while the CVAE predicts timing and reproduces held-out feature distributions more closely. These statements are paired at the participant level and survive correction.

Endpoint	CVAE	Spline+PCA	CVAE better	BH-FDR p	Endpoint	CVAE	Spline+PCA	CVAE better	BH-FDR p
Trajectory MSE	0.177	0.097	6/28	3.3e-05	Trajectory MSE	0.032	0.023	4/28	8.3e-04
Initiation MAE	37.8	56.2	27/28	1.2e-07	Initiation MAE	34.8	55.6	27/28	1.2e-07
Movement MAE	43.4	51.2	24/28	9.9e-05	Movement MAE	44.2	50.2	23/28	0.0014
Mean KS	0.278	0.325	26/28	1.6e-06	Mean KS	0.219	0.255	23/28	1.1e-04
MMD	0.141	0.189	24/28	5.8e-05	MMD	0.074	0.132	26/28	3.1e-06

Table 2a. $n=3$ paired comparisonTable 2b. $n=8$ paired comparison

Figure 3. CVAE capacity and fingerprint enrollment

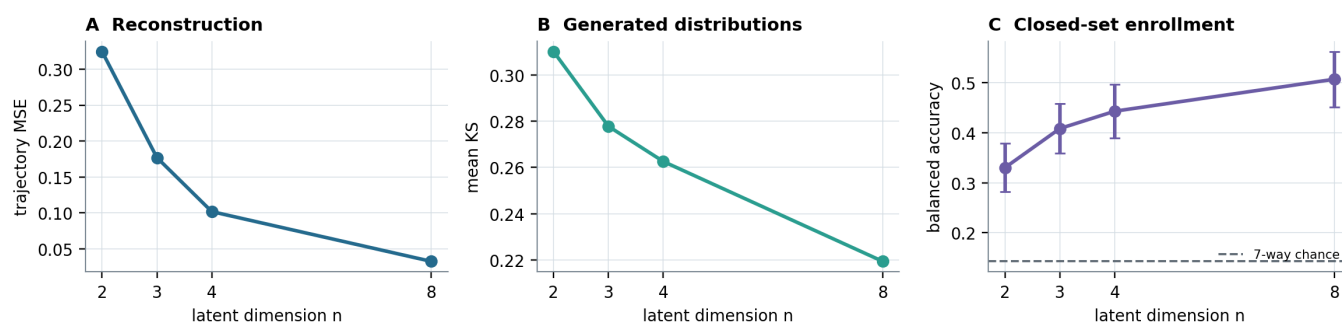


Figure 3. Increasing latent capacity improves CVAE reconstruction, feature-distribution fidelity, and descriptive enrollment accuracy. The $n=3$ model preserves the requested low-dimensional interface; $n=8$ is the stronger capacity benchmark. The current data do not support claiming that $n=3$ is sufficient for the complete generative objective.

5.2 What the condition-only baseline establishes

CVAE $n=3$ improves trajectory and timing prediction over condition-only Ridge for every participant, but not mean KS or MMD. CVAE $n=8$ improves all primary endpoints for most participants (BH-FDR $p < 0.001$ for MSE, timing, mean KS, and MMD). Thus low-dimensional latent codes carry participant-specific predictive information, while strong distribution-level benefit over task metadata appears only at higher capacity.

6. Conditioning and latent interpretation

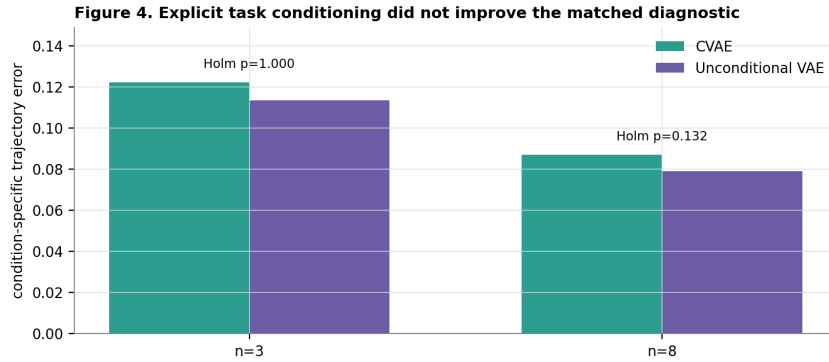


Figure 4. With the context fingerprint and latent draws fixed, CVAE and unconditional-VAE condition-stratum trajectory errors do not differ significantly ($n=3$ Holm $p=1.000$; $n=8$ $p=0.132$). Across 44 feature-level tests, no conditioning advantage survives BH-FDR. Task controls are therefore architecturally available but not validated as participant-specific causal controls. Axis-specific latent correlations are shown only as run-specific dashboard diagnostics because independent VAE fits are sign-, permutation-, and rotation-indeterminate.

7. Minimum-jerk results

Figure 5. Minimum-jerk component analysis is model-order sensitive

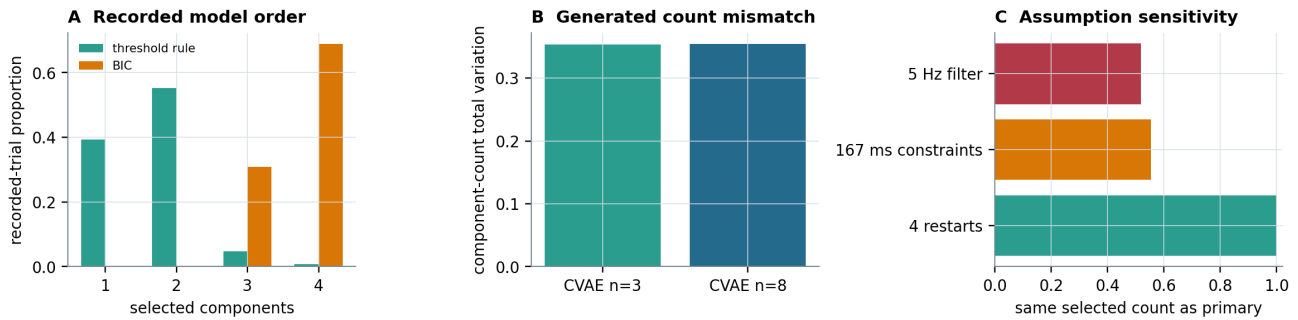


Figure 5. Recorded component counts depend strongly on model-order criterion. Generated count mismatch remains substantial (total variation 0.353 for $n=3$; 0.355 for $n=8$), and none of nine $n=3$ versus $n=8$ component-fidelity contrasts survives BH-FDR. Additional optimizer restarts preserve the threshold count on 100% of 56 sampled trials, whereas 167 ms constraints and a 5 Hz filter preserve it on only 55.4% and 51.8%. The decomposition supports kinematic description but not a cognitive-strategy label without Prof. Friedman's validation of filter and order rules.

8. Discussion and lessons learned

Supported	Boundary
Participant structure is non-random and context enrollment is above nominal chance.	Enrollment uses trials from the test participant; it is not zero-shot identity recognition.
Spline+PCA is the best deterministic shape compressor at matched n.	This does not make it the best stochastic generator or timing model.
CVAE n=8 improves timing and distribution fidelity over spline and task-only Ridge.	Unconditional VAE is statistically comparable; explicit condition value is unproven.
CVAE n=3 is a usable compact fingerprint interface.	Its distribution advantage over the condition-only baseline is not established.
Minimum-jerk parameters offer interpretable kinematic summaries.	Component count is analysis-dependent and cannot currently be called strategy.

8.1 Timing-head failure mode

Log-timing heads yield median initiation errors near 22-27 ms and 95th-percentile errors near 99-117 ms, but isolated extrapolations reach 11.5 s for CVAE n=8. These rare failures can make R^2 strongly negative despite moderate MAE. Participant-balanced MAE is therefore the primary timing endpoint; the unclipped outliers are disclosed in Figure A2 rather than hidden by post-hoc bounds.

8.2 Limitations

The cohort contains 28 participants and no independent acquisition cohort. Context enrollment is required for unseen participants. Phase normalization intentionally separates shape from time but can remove local velocity information. Generated distributions are evaluated on selected kinematic summaries and cannot certify perceptual realism. Latent axes are not identifiable across independent fits. The minimum-jerk order rule, 10 Hz filter, short-task temporal bounds, late-trial threshold, and condition-2 fixation flag require domain confirmation.

8.3 End product and future work

The Streamlit dashboard exposes the strategy-window generator, context fingerprints, n=3/n=8 models, held-out metrics, and run-specific latent diagnostics. Condition sliders are labeled exploratory. After Prof. Friedman's review, the priority is to freeze task-specific exclusions and minimum-jerk settings, rerun only affected stages, calibrate or bound timing outputs in a pre-specified manner, and evaluate an independent cohort if available. A hierarchical subject-level VAE is a publication-oriented extension, not required for the workshop result.

References

- [1] Brenner, E. and Smeets, J.B.J. Continuously updating one's predictions underlies successful interception. *J Neurophysiol* 120, 3257-3274 (2018). doi:10.1152/jn.00517.2018.
- [2] Slowinski, P. et al. Dynamic similarity promotes interpersonal coordination in joint action. *J R Soc Interface* 13, 20151093 (2016). doi:10.1098/rsif.2015.1093.
- [3] Kingma, D.P. and Welling, M. Auto-Encoding Variational Bayes. ICLR (2014). arXiv:1312.6114.
- [4] Sohn, K., Lee, H. and Yan, X. Learning Structured Output Representation using Deep Conditional Generative Models. NeurIPS 28 (2015).
- [5] Flash, T. and Hogan, N. The coordination of arm movements: an experimentally confirmed mathematical model. *J Neurosci* 5, 1688-1703 (1985). doi:10.1523/JNEUROSCI.05-07-01688.1985.
- [6] Friedman, J. *submovements* repository, commit 9c2f40c. github.com/JasonFriedman/submovements.

Appendix A. Representative held-out outputs

Figure A1. Held-out reconstruction and context-fingerprint generation (fold 0, seed 42)

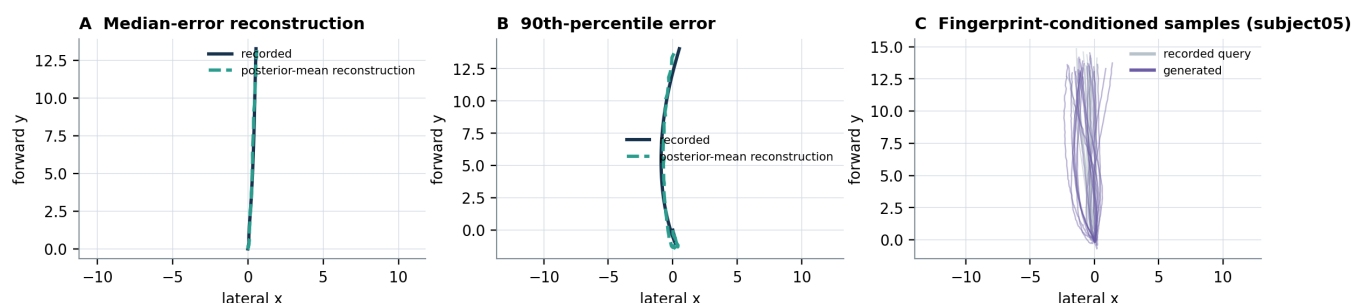


Figure A1. Posterior-mean reconstructions and fingerprint-conditioned generation for a held-out participant. Reference run: fold 0, CVAE $n=8$, seed 42; typical trial subject13_li_2_1_2_7; high-error trial subject19_li_2_1_1_8. This figure is illustrative; inferential results use every fold and seed.

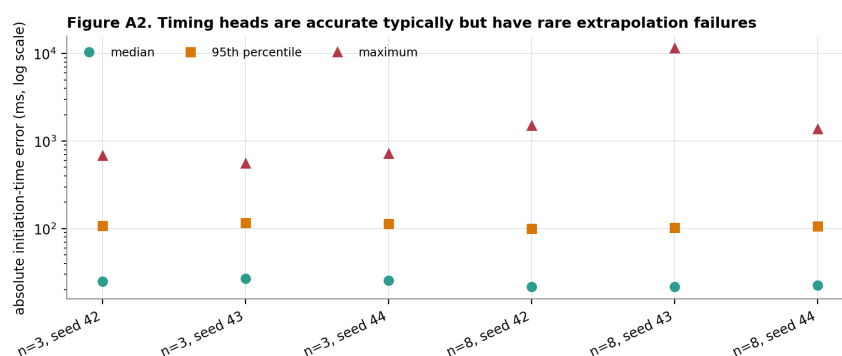


Figure A2. CVAE initiation-time error distribution by seed. Log scale makes the separation between typical and worst-case behavior visible. Full raw and seed-averaged tables, feature-wise fidelity files, condition diagnostics, and minimum-jerk sensitivity outputs accompany the executable repository.

Questions for Prof. Friedman

Confirm (i) whether condition-2 'Not fixating on the dot enough!!!' trials should remain; (ii) the intended late-trial reference event and threshold; (iii) whether 10 Hz and our short-task component bounds are appropriate; and (iv) the preferred minimum-jerk model-order criterion before strategy-level interpretation.